

IUSUP ABDULIAPAROV

C# Simulation Software Engineer | Real-Time 3D, Rendering & Physics

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PROFESSIONAL SUMMARY

C# simulation software engineer specialising in real-time medical imaging, numerical transport, instrument physics and performance-critical 3D systems. Sole engineer on a coronary angiography trainer validated against reference angiograms with clinical reviewers. Earlier work reached 9M+ users at 40k DAU peak, reducing a WebGL build from 100 MB to 20 MB and improving low-end Android performance from ~15 to a stable 60 FPS.

WORK EXPERIENCE

Coronary Angiography Simulator Simulation Software Engineer

2025 to Present

Sole engineer on a coronary angiography trainer spanning X-ray imaging, contrast transport, instrument physics and balloon/stent treatment, validated against reference angiograms with clinical reviewers.

- Built a five-pass URP fluoroscopy pipeline with tissue-density accumulation, detector gain/lag, DSA road-mapping and a CT-volume path; tuned frame by frame against reference angiograms.
- Implemented advection-diffusion contrast transport on a 19-segment vessel graph derived from the artery mesh, reproducing anatomical filling, stenosis slowdown and occlusion.
- Reduced the injection frame hitch 2.6x, from 12.3 ms to 4.7 ms, while physics already consumed 59% of the frame.
- Made multiple boluses superpose in one field while the emitting volume tracks the catheter tip through a breathing, beating vessel.
- Built tail-driven rod-solver physics with lumen coupling for coaxial catheter/guidewire motion; configured eight instrument profiles without code.
- Simulated treatment mechanics: balloon dilation with elastic recoil, plus a stent that expands, implants on deflation and stays fixed to the breathing vessel wall.

Tech: Unity, C#, URP, ShaderLab/HLSL, Burst, custom render features, ScriptableObjects, procedural meshes.

Aviana Studios, WebGL and Mobile Products

Remote

Real-Time 3D Software Engineer & Co-founder

2023 to 2026

Owned C# and real-time 3D engineering for a high-traffic WebGL product reaching 9M+ users at 40k DAU peak, plus nine standalone WebGL titles built on a shared codebase.

- Shipped nine standalone WebGL titles from a shared C# codebase with reusable services, state machine and save layer across per-game feature branches.
- Reduced WebGL build size from 100 MB to 20 MB through shader and asset scoping plus media re-encoding; improved low-end Android from ~15 FPS to stable 60 FPS through GC reduction, pooling, batching and profiler-driven optimization.
- Built a 1k-line custom Rigidbody vehicle controller: spring-damper suspension with bump stops and sphere-cast wheel probes, drift, boost, downforce, air control and auto-recovery; refactored from an earlier ConfigurableJoint prototype to stabilize handling for the game's arcade feel.
- Designed reusable architecture with dependency injection, hierarchical state machines, ScriptableObject configuration, reactive UI and a multi-backend persistence layer.
- Optimized environment geometry from ~1M to ~40k triangles on a mobile prototype via mesh decimation, LODs and silhouette-preserving proxies; supported production releases across WebGL and mobile platforms under strict performance, memory and build-size constraints.

Tech: Unity, C#, URP, VContainer, R3, UniTask, Addressables, WebGL, Android.

TECHNICAL SKILLS

- **Architecture:** SOLID, dependency injection (VContainer, custom service locator), hierarchical state machines, ScriptableObject configuration, reactive systems (R3, UniRx, UniTask), multi-backend persistence
- **Numerical Simulation:** advection-diffusion transport, vessel-graph solvers, stenosis and occlusion behaviour, multi-bolus fields, rod and soft-body physics
- **Rendering & Medical Imaging:** custom URP render features, five-pass fluoroscopy, DSA road-mapping, density accumulation, detector gain and lag, CT-volume path
- **Physics & Real-Time 3D:** catheter and guidewire lumen coupling, custom Rigidbody vehicle controller, ConfigurableJoints, procedural meshes, skinned deformation
- **Performance:** Unity Profiler, Memory Profiler, GC reduction, draw-call batching, WebGL build-size optimization, low-end Android profiling

EDUCATION

Università Cattolica del Sacro Cuore

BSc Economics & Management

Ufa State Petroleum Tech. University

Automation & Control Systems

Milan, Italy

2024 to Present

Ufa, Russia

2022 to 2024